

Thawsif Chowdhury

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EDUCATION

Lehigh University, Bethlehem, PA

Graduation: May 2029

Bachelor of Science in Electrical Engineering

GPA: 3.96

Related Coursework: Circuit Analysis, Electromagnetics, Computer Engineering Fundamentals

Personal Projects

Automated Irrigation System

June 2024 – July 2024

- Designed a controlled irrigation system using Arduino UNO for maintaining optimum soil moisture levels
- Programmed in C++ to display real-time analog data of moisture levels using a soil sensor
- Implemented a threshold logic (500 ADC) which activated the motor via a relay module to water plants

Mechanical Arm

March 2024 – April 2024

- Designed a 5-fingered mechanical hand controlled using an Arduino UNO
- Implemented servo motors with strings that connected to the fingers
- Programmed the control software in Arduino IDE (C++) to rotate servo motors in response to keyboard inputs for finger articulation

EXPERIENCE

IEEE Student Branch Treasurer

September 2025 – Present

Institute of Electrical and Electronics Engineers, Lehigh University, Bethlehem, PA

- Develop annual calendar year branch funding to organize 5+ major events for 50+ members

Cybersecurity Intern

August 2026 – Present

BVA Bear's IT Solutions - Virtual

- Review 15+ policies regarding AUP, SOPs to ensure company meets CMMC Level 1 Certification

Software Engineering Intern

June 2026 – August 2026

ThinkNeuro - Virtual

- Managed version control using Git fundamentals such as merging, pulling, branching, committing to prevent code conflicts among collaborators
- Deployed an open-source website built with Astro and Tailwind CSS to highlight personal projects
- Developed a multithreaded object detection system using YOLO11 and OpenCV to track real-time dimensions and directions of 35 classes, triggering text-to-speech alerts, by training on 10k+ images

Global Social Impact Fellow - PureSpaces

January 2026 – Present

Lehigh Creative Inquiry, Bethlehem, PA & Almaty, Kazakhstan

- Led the experimental validation of PureAura filtration systems, executing 3 trials using incense, IQ Air monitors, and purifiers to analyze 1-minute interval CSV data, ensuring 60% faster PM 2.5 recovery
- Developed a monitoring program requiring zero manual intervention since deployment to ingest 35,000 hourly PM 2.5 readings every 12 months across 4 testing sites, by integrating the Almaty Air Initiative API and configuring a cloud-based cron scheduler to execute the task

America Reads America Counts Tutor

September 2025 – May 2026

Lehigh Community Service Office, Bethlehem, PA

- Tutored 40+ 8th-grade students in Science, English, Math to prepare them for high school coursework
- Dedicated 3 hours per week across 3 sessions to clarify 10+ core concepts in Science, English, Math

Engineering Student Researcher

July 2024 – August 2024

Vaughn College, Flushing, NY

- Engineered and trained a Convolutional Neural Network (CNN) model for handwritten digit recognition and a Long Short-Term Memory (LSTM) model for performing sentiment analysis on a dataset of 50,000 IMDb movie reviews
- Implemented a machine learning system using a Linear Regression model for analyzing housing prices
- Engineered an AI-powered voice recognition program using Python to control game movements

SKILLS

Programming Languages: Python, Java (Basics), Arduino (C++), MATLAB

Technical Skills: Arduino UNO, Tinkercad, LTSpice, Git, GitHub, MS Office Suite, Google Suite